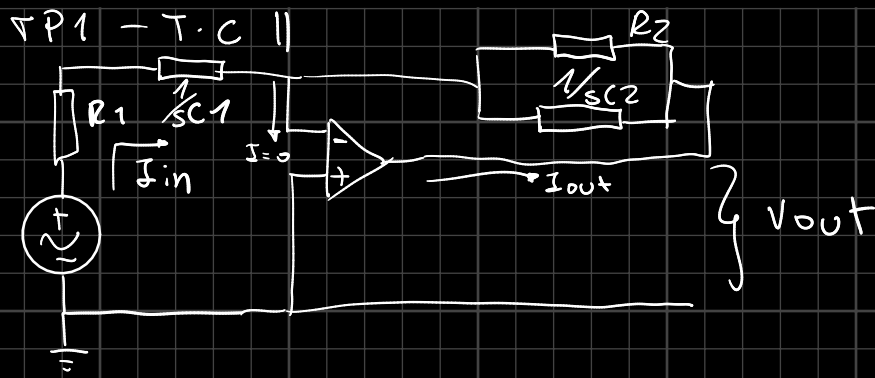




$$Z_i = R_1 + \frac{1}{sC_1}$$

$$Z_o = \left(\frac{1}{R_2} + sC_2 \right)^{-1}$$

$$Z_i = \frac{sR_1 \cdot C_1 + 1}{sC_1}$$

$$Z_o = \frac{R_2}{sC_2 R_2 + 1}$$

$$H(s) = \frac{V_o}{V_i}$$

$$\frac{V_{out}}{V_{in}} = H(s) = -\frac{Z_{out}}{Z_{in}}$$

$$I_{in} = I_{out}$$

$$H(s) = -\left(\frac{R_2}{sC_2 \cdot R_2 + 1} \right) \cdot \left(\frac{sR_1 \cdot C_1 + 1}{sC_1} \right)$$

$$\frac{V_{in} - V_-}{Z_{in}} = \frac{V_- - V_{out}}{Z_{out}}$$

$$H(s) = \frac{-sR_2 \cdot C_1}{(sC_2 R_2)(sR_1 C_1) + (s \cdot C_2 \cdot R_2) + (sR_1 \cdot C_1) + 1}$$

$$\frac{V_{in}}{Z_{in}} = -\frac{V_{out}}{Z_{out}}$$

$$= \frac{-sR_2 \cdot C_1}{s^2 C_2 R_2 R_1 C_1 + s(C_2 R_2 + R_1 \cdot C_1) + 1}$$